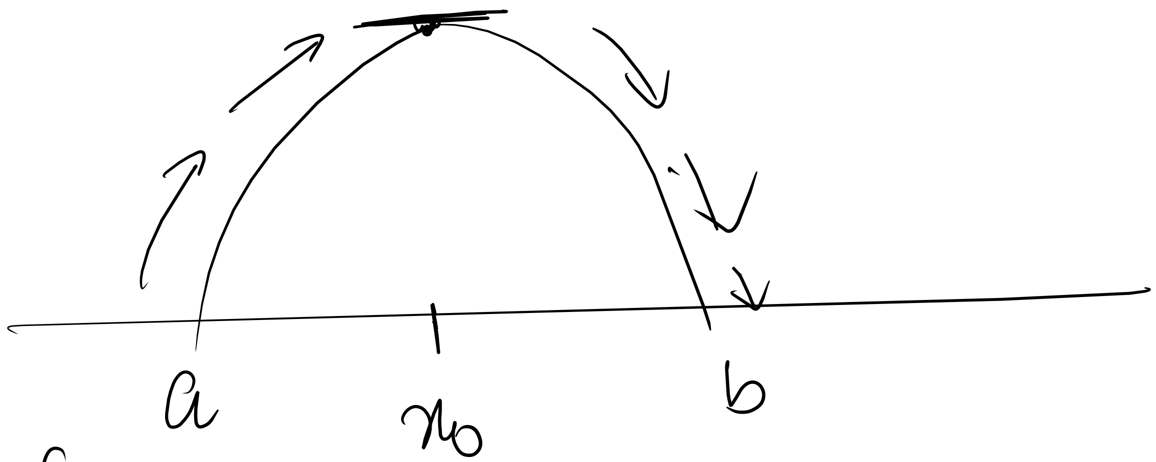
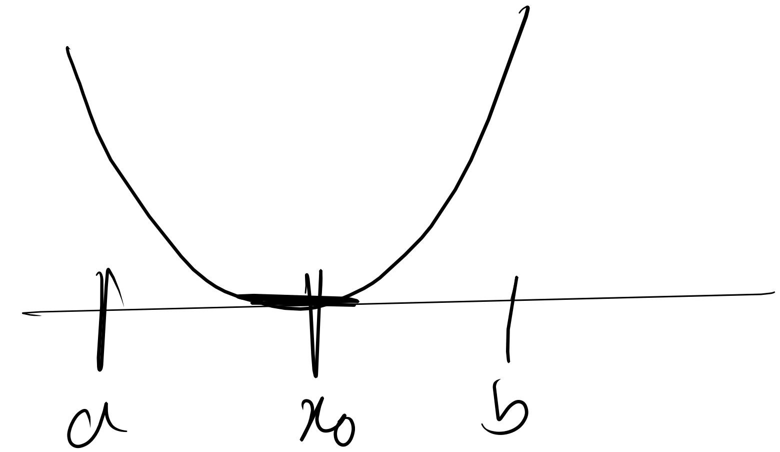
First Derivative Test :





(i) If $f'(x) > 0$ on (a, x_0) & $f'(x) < 0$ on (x_0, b), f has a maximum at x_0

(ii) If $f'(x) < 0$ on (a, x_0) & $f'(x) > 0$ on (x_0, b), f has a minimum at x_0.

(iii) If $f'(x)$ has same sign on both (a, x_0) & (x_0, b) f doesnot have extremum (max/min) at x_0.

Ex: $f(x) = 2x^3 - 9x^2 + 12x$

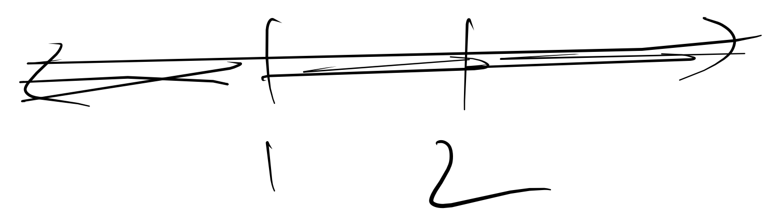
Stationary points (Critical points)

$$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2).$$

$f'(x) = 0$ ⇒ Verify

$$\Rightarrow 6(x-2)(x-1) = 0$$

$x = 2, 1$



Interval	Testpoint	sign of $x-2$	sign of $x-1$	sign of $f'(x)$
$x<1$	0	$-$	$-$	$+$
$1<x<2$	$3/2$	$-$	$+$	$-$
$x>2$	3	$+$	$+$	$+$

(i) $f'(x)>0$ on $(-\infty, 1)$. & $f'(x)<0$ on $(1, 2)$. f has a relative maximum at 1. $f(1)=2-9+12=5$. (max)

(ii) $f'(x)<0$ on $(1, 2)$ & $f'(x)>0$ on $(2, \infty)$

f has relative minimum at 2.

$f(2)=2.(2)^3-9(2)^2+12.2=4$

$$f(x) = x^4 - 8x^3 + 22x^2 - 24x + 5$$

$$f'(x) = 4x^3 - 24x^2 + 44x - 24$$

$$f'(x) = 0$$

$$\Rightarrow 4x^3 - 24x^2 + 44x - 24 = 0$$

$$\Rightarrow 4(x^3 - 6x^2 + 11x - 6) = 0$$

$$\Rightarrow x^3 - 6x^2 + 11x - 6 = 0$$

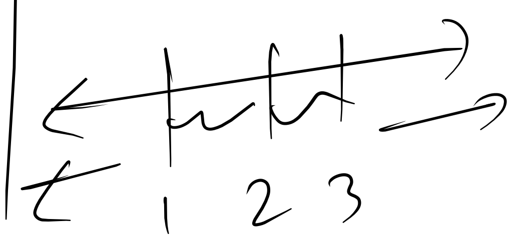
$$\Rightarrow x^3 - x^2 - 5x^2 + 5x + 6x - 6 = 0$$

$$\Rightarrow x^2(x-1) - 5x(x-1) + 6(x-1) = 0$$

$$\Rightarrow (x^2 - 5x + 6)(x-1) = 0$$

$$(x-1)(x-2)(x-3) = 0$$

$$x = 1, 2, 3$$



H.W

1. Outline Sheets

2. H.W.

3. Practice Sheets

Solve

Exam

for

H.W.

$$f(x) = 17 - 15x + 9x^2 - x^3$$